

MICHAEL RIZZO SMITH

michael.smith.4@vanderbilt.edu | michael.josmith21@gmail.com | mjosmith.github.io

EDUCATION

Vanderbilt University

Ph.D Astrophysics

Advisor: Kelly Holley-Bockleemann

Nashville, TN

Aug 2023 - Exp. 2029

Stony Brook University

Bachelor of Science, Physics and Astronomy & Planetary Science

Magna Cum Laude with Honors in Physics

Stony Brook, NY

Aug 2017 - May 2021

AWARDS AND FELLOWSHIPS

Award Recipient, NSF Graduate Research Fellow

Jun 2025

Award Recipient, Vanderbilt Graduate School Travel Grant

July 2024

Award Recipient, McMinn Research Award

May 2024

Award Recipient, Vanderbilt Robert T. Lagemann Award

Apr 2024

Departmental Honors, Dept. of Physics

May 2021

Award Recipient, Stony Brook Campus Scholarship

Aug 2017 - May 2021

PROFESSIONAL WORK AND TEACHING

Teaching Assistant, ASTR 1010L: Intro to Astronomy Lab

Fall 2023 - Spring 2025

Co-Coordinator, Vanderbilt's Graduate Student Astrophysics Journal Club

Aug 2024 - present

Code Captain, LISA, Disc-IMRI Astrophysics Working Group

Dec 2023 - present

Member, Establishing Multimessenger astronomy Inclusive Training (EMIT)

Sep 2023 - present

Data Analyst, The Ohio State University

Jun 2021 - July 2023

PRESENTATIONS

Invited Speaker, EMIT Outreach

Jan 2025

Presented to a local public high school on computational astrophysics, use of simulations in science, and different pathways to academia.

Talk, Vanderbilt's Astrophysics Journal Research Club

Jan 2025

The dynamics and electromagnetic signatures of accretion in unequal mass binary black hole inspirals

Talk, The 15th International LISA Symposium

July 2024

Presenting on behalf of the Disc-IMRI Code Comparison LISA Astrophysics Working Group

Invited Speaker, Valor Collegiate Academy

Mar 2024

Invited to speak to first-year class of a local public high school on a career in Astrophysics and my personal path.

Talk, Vanderbilt's Astrophysics Journal Research Club

Feb 2024

Astrophysical Disk Modeling With Hydrodynamic Simulations

PUBLICATIONS

Accepted or submitted to peer-reviewed scientific journals

(1 first-author, 3 co-author, 92 citations)

1. J. M. M. Neustadt, C. S. Kochanek, and **Rizzo Smith, M.** *Constraints on pre-SN outbursts from the progenitor of SN 2023ixf using the large binocular telescope.* In: MNRAS 527.3 (Jan. 2024), pp. 5366–5373. arXiv: 2306.06162 [astro-ph.HE]
2. Matthew Kenworthy, [20 authors], and **Rizzo Smith, Michael.** *A planetary collision afterglow and transit of the resultant debris cloud.* In: Nature 622.7982 (Oct. 2023), pp. 251–254. arXiv: 2310.08360 [astro-ph.EP]
3. **Rizzo Smith, M.**, C. S. Kochanek, and J. M. M. Neustadt. *The late time optical evolution of twelve core-collapse supernovae: detection of normal stellar winds.* In: MNRAS 523.1 (July 2023), pp. 1474–1495. arXiv: 2212.09763 [astro-ph.HE]
4. A. Kawash, [11 authors], **Rizzo Smith, M.**, T. W. -S. Holoien, J. L. Prieto, and T. A. Thompson. *The Galactic Nova Rate: Estimates from the ASAS-SN and Gaia Surveys.* In: The Astrophysical Journal 937.2, 64 (Oct. 2022), p. 64. arXiv: 2206.14132

Non-Referred Publications

1. M. Rizzo Smith and ASASSN-Team. *An Update on ASASSN-21qj: A Rapidly Fading, Sun-Like Star; Back With a Vengeance.* In: The Astronomer's Telegram 15531 (July 2022), p. 1
2. M. Rizzo Smith and ASASSN-Team. *ASASSN-22el: A Deep Eclipse Event.* In: The Astronomer's Telegram 15308 (Apr. 2022), p. 1
3. M. Rizzo Smith and ASASSN-Team. *ASASSN-21sa: A Deep Dimming Event.* In: The Astronomer's Telegram 14937 (Sept. 2021), p. 1
4. M. Rizzo Smith and ASASSN-Team. *ASASSN-21qj: A Rapidly Fading, Sun-Like Star.* In: The Astronomer's Telegram 14879 (Aug. 2021), p. 1
5. M. Rizzo Smith and ASASSN-Team. *ASASSN-21nn: An Unusual Dimming Event in a Red Giant Star.* In: The Astronomer's Telegram 14803 (July 2021), p. 1
6. M. Rizzo Smith and ASASSN-Team. *ASASSN-21ml: Discovery of an Extreme ($\Delta g > 10$ mag) L-Dwarf Flare.* In: The Astronomer's Telegram 14778 (July 2021), p. 1